

# The Importance of Sensory Feedback to Enhance Embodiment During Virtual Training of Myoelectric Prostheses Users

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## ABSTRACT

In this poster, we propose a system that uses immersive Virtual Reality (iVR) and EMG signal processing (muscle activity) to provide a training environment for amputees who are supposed to use a myoelectric prosthesis. We also investigate the efficiency of learning how to control a virtual prosthesis with and without sensory feedback. Our results show that virtual training can be greatly improved when proper tactile feedback is provided, especially for myoelectric controlled prostheses.

**Keywords:** Immersive Virtual Reality, myoelectric prosthesis, prosthetic training, sensory feedback.

**Index Terms:** K.4.2. [Computers and Society]: Social Issues-Assistive technologies for persons with disabilities.

## 1 INTRODUCTION

In the past years, Virtual Reality (VR) has been considered a potential approach for prosthetic training [1]. However, most VR-based upper-limb prosthetic training systems, reported in the literature, are based on non-immersive environments [2, 3]. In fact, in many initiatives, users only visualize a virtual prosthesis, at a distance, on a flat-screen. Unfortunately, this lacks intuition for real life prosthetic control.

Furthermore, sole visual feedback drastically limits the immersive experience for the purpose of prosthesis training, which requires users to reach and manipulate different objects. It is important to provide users with the feeling of touch to enhance the experience and potentially provide sensations usually present when training with real devices [4]. Such feedback is especially crucial for myoelectric prostheses.

This paper raises the argument that an immersive virtual environment equipped with proper sensory feedback can significantly lessen the above drawbacks. We hypothesize that such integration can provide the user with a high immersion level combined with improved prosthesis embodiment in the virtual space.

## 2 MATERIAL AND METHODS

### 2.1 System Architecture

To evaluate this proposal, an iVR system has been designed. The system's main components are: HTC Vive Pro, Vive Tracker, EMG

controller (data acquisition and pattern recognition), vibrational armband, and a computer (Fig. 1). The user interacts with the virtual environment by moving and contracting the stump muscles. In case of nonamputee volunteers, forearm muscles are used.

The Vive Tracker, located in the arm, provides information so that the virtual prosthesis moves within the virtual environment as the user moves his/her arm. The EMG module (Fig. 1) processes the contractions of the stump (or forearm) muscles and sends setpoint values to the virtual environment to indicate how much the virtual prosthesis must open/close the hand. The vibrational armband, and the EMG sensors are also positioned on the user's forearm. Whenever the user grabs an object in the virtual environment, vibration feedback is sent to the user to convey touch. If the object is dropped or released, the feedback stops.

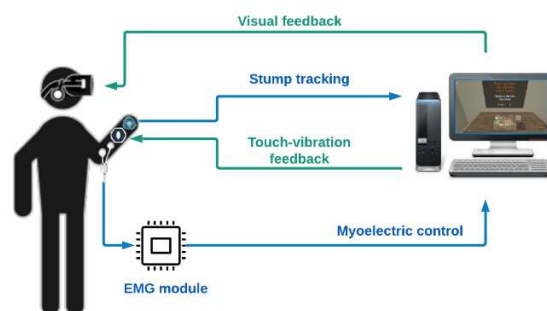


Figure 1: System Architecture.

### 2.2 Virtual Training Environment

A virtual environment was designed to simulate the Box and Blocks Test (B&B), one of the most used measures for evaluating functionality and users' performance [6]. The subject must use the device to grasp wooden blocks and move them, one at a time, over a barrier. Figure 2 shows the virtual training environment with the virtual prosthesis in the bottom right-hand side.

The size and the number of blocks can be modified, according to a desired protocol. Time to move and number of blocks are metrics.



Figure 2: iVR environment for the Box and Blocks Test.

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### 2.3 Virtual Prosthesis

In this work, users had to complete B&B tasks using one of two grasp types: (A) power grasp and (B) pinch. Both grasps can be adjusted from fully opened (0% closed) to fully closed (100% closed). The setpoint values are sent to the virtual environment by the EMG module to reflect the subject's movement intention. A 3D printed version of the prosthesis was produced and tested to evaluate its functionalities and ensure that the virtual prosthesis was correctly mimicking the real one. The 3D printed prosthesis was also used to ensure that the virtual B&B tasks could be performed as required.

### 2.4 Participants

For this study, eight non-amputee male volunteers, ages between 20 and 40, were recruited. These volunteers were divided into two groups: G1 - performed the B&B tasks while receiving visual and vibrational tactile feedback; G2 - performed the B&B tasks while receiving only visual feedback. The experiment should be carried out with more volunteers, but due to the COVID-19 pandemic crises and the availability of volunteers, this was not possible.

### 3 EXPERIMENTAL PROCEDURE

Before the experiment, the volunteers received instructions related to the task, how they should use muscle contractions to control the virtual prosthesis, and the type of feedback to be expected in the case of G1. The skin was prepared, and EMG electrodes were positioned over the relevant muscles on the forearm. The reference electrode was placed on the contralateral wrist, which was still during the tasks, to minimize motion artifacts. For calibration, the user must rest the forearm muscles for 5 seconds and contract for another 5 seconds at a comfortable level.

The Vive Tracker and the vibrational armband were positioned side-by-side close to the wrist. A 60 seconds preliminary run was allowed so that the volunteer could get used to the virtual environment and make sure he could control the virtual prosthesis using EMG signals. Finally, the volunteer was donned with the Vive headset. The task consisted four B&B trials. In each trial the volunteers should move all blocks from one side of the box to the other as fast as possible. The time to complete each trial was registered.

### 4 RESULTS AND DISCUSSION

In the first sessions, the volunteers learn how to control the virtual prosthesis using myoelectric signals and adapting to the virtual environment. Accordingly, the first two trials required a longer time to complete. As the trials progressed, especially for the volunteers of G1, completion time decreased steadily. Completion time for the volunteers of G1 during Trial 1 vary from 17.67 to 209.97 seconds (mean 100.99 seconds); completion time for Trial 2 ranged from 25.39 to 82.32 seconds (mean 47.6 seconds); for Trial 3 from 17.1 to 62.22 seconds (mean 36.75 seconds); and for Trial 4 from 17.59 to 34.25 seconds (mean 22.21).

In contrast, the completion time for the volunteers of G2 during Trial 1 vary from 28.06 to 135.12 seconds (mean 66.25 seconds); completion time for Trial 2 ranged from 10.4 to 207.38 seconds (mean 85.77 seconds); for Trial 3 from 11.81 to 42.67 seconds (mean 32.49 seconds); and for Trial 4 from 13.94 to 96.36 seconds (mean 50.87).

The mean completion time for each group per trial and the standard deviation can be seen in Figure 3, where the tendency line, for each trial, can be seen in red (G1) and blue (G2). As it can be noted, although the initial mean value for G1 was considerably higher than the initial mean value for G2 (mainly due to the very high time taken by G1-V3 to complete the first trial), the group

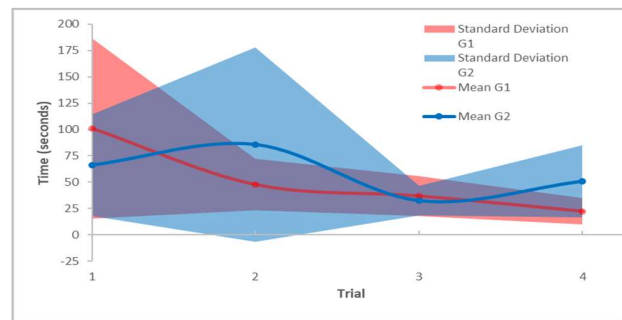


Figure 3: The mean completion time for each trial.

using vibrational tactile feedback showed a better performance overall. On average, G2 took 2.3 times longer to complete the task at the fourth trial.

### 5 CONCLUSION

In this paper, we investigated the importance of sensory feedback to enhance embodiment during myoelectric prostheses using iVR. We developed an immersive virtual environment where the users can perform the tasks required by Box & Blocks Test using a virtual myoelectric prosthesis and vibration feedback.

Although both experimental groups showed improvement in the control of the myoelectric virtual prosthesis over the trials, participants of the group using tactile vibrational feedback showed better performance as the trials progressed, compared with the group that did not use tactile feedback. Those results indicate that sensory feedback is indeed capable of providing a better overall experience while optimizing training sessions using immersive virtual reality environments.

As soon as the current COVID-19 pandemic crises subsides, future steps for this work involve extensive tests with amputee subjects seeking to perfect the system with that population.

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